

Project Title

Smart Ground-Truthing and Digital Biodiversity System for Plant Species Documentation

Industry partner

NeuronAI (Sarawak Forestry Corporation's commercialisation partner)

Introduction

Accurate plant species documentation is fundamental to biodiversity conservation, ecological research, and sustainable ecotourism. However, current field documentation practices in Sarawak's protected areas remain largely manual, requiring botanists and conservation officers to identify plants, attach physical labels, record observations on paper, and later transfer the information into digital databases. This process is time-consuming, labour-intensive, susceptible to data loss, and difficult to maintain consistently across large forested landscapes.

To address these challenges, Sarawak Forestry Corporation (SFC), together with Swinburne University of Technology Sarawak, is developing a Smart Ground-Truthing and Digital Biodiversity System for Niah National Park. The system combines QR-based plant tagging, an offline-capable mobile application, GPS-assisted location tracking, and a centralized biodiversity database to streamline field data collection and management. Users can scan QR codes attached to tagged plants, record species information, upload photographs, capture GPS coordinates, and synchronize collected data with a cloud database whenever internet connectivity becomes available.

The platform also supports researchers, conservation officers, and park visitors through a digital plant knowledge system that provides taxonomic information, ecological characteristics, conservation status, and cultural significance of plant species. QR-enabled educational content can further enhance visitor experiences and community-based ecotourism while promoting biodiversity awareness.

This integrated digital solution aims to modernize biodiversity documentation, improve conservation management, strengthen ecological research, and enhance sustainable ecotourism in Sarawak's protected forests.

Problem Statements:

1. Manual and Inefficient Biodiversity Documentation

Current plant documentation relies on handwritten records, physical tags, and manual data entry, resulting in duplicated work, inconsistent records, inefficient field operations, and increased risk of information loss.

2. Limited Digital Accessibility and Knowledge Sharing

Researchers, conservation officers, local communities, and visitors have limited access to up-to-date biodiversity information, reducing the effectiveness of conservation management, scientific research, and knowledge-based ecotourism.

Objectives:

- **Objective 1: Develop a Smart Digital Ground-Truthing Platform**
Develop an integrated mobile and web-based biodiversity documentation platform that enables QR code scanning, plant species recording, offline data collection, GPS-based location tracking, and synchronization with a centralized biodiversity database.
- **Objective 2: Develop a Digital Plant Knowledge System**
Create a searchable digital knowledge repository containing plant taxonomy, ecological information, conservation status, photographs, and cultural significance to support researchers, conservation officers, park guides, and visitors.
- **Objective 3: Ensure Secure Biodiversity Data Management**
Implement secure authentication, encrypted data storage, role-based access control, and cloud synchronization to protect biodiversity records and user information.
- **Objective 4: Implement IoT-Based Plant Protection**
Develop an IoT-enabled monitoring system that collects real-time environmental and location data, detects potential threats, and supports rapid conservation response for protected plant species.
- **Objective 5 (Optional): Develop an Innovative Ground-Truthing Workflow for Biodiversity Documentation**
Design and develop an innovative approach to improve the efficiency, accuracy, and scalability of plant ground-truthing and biodiversity data collection. The proposed solution should address the limitations of current approaches, such as manual recording and physical tagging, particularly in remote forest environments with limited connectivity and challenging field conditions.

Areas of Focus (Must Implement ALL):

1. Smart Digital Ground-Truthing Platform

- **Implementation:** Develop a *mobile* application for biodiversity documentation.
- **Features:**
 - QR code scanning for tagged plant species
 - Plant species database (searchable plant information and identification reference)
 - Offline field data collection
 - GPS location recording
 - Cloud synchronization with a central database
- **Outcome:** Botanists and conservation officers can efficiently identify, document, and manage plant species information during field surveys, reducing manual effort and improving data quality.
- **Basic functionalities:**
 - Botanist Role
 - Register new plant records on site

- Record detailed plant information (e.g., taxonomy, morphological characteristics, height, and related attributes).
- Capture and attach photographs of the plant
- Record its GPS location
- Synchronize field data with the central database
- Generate a QR code for each plant record, allowing users to view the recorded plant details.

2. Digital Plant Knowledge System

- **Implementation:** Develop a web-based digital biodiversity knowledge system for managing and accessing plant species information.
- **Features:**
 - Plant taxonomy and scientific names
 - Species descriptions
 - Botanical photographs
 - Conservation status
 - Species distribution maps
 - Search and filtering by scientific or common name
- **Outcome:** Conservation officers can efficiently manage, retrieve, and update accurate plant biodiversity information to support conservation, research, and public education.
- **Basic functionalities:**
 - Conservation Officer Role
 - Add, edit, and delete plant species records
 - Upload plant photographs and supporting information
 - Review and approve submitted observations
 - Search and filter biodiversity records
 - Generate and export biodiversity reports

3. Cybersecurity and Data Protection

- **Implementation:**
 - Secure login for all users.
 - Encryption for personal data, performance records, and assessments.
 - Access control for admins, trainers, and guides.
- **Outcome:** Protect sensitive information, ensuring compliance with data protection regulations and maintaining trust in the platform.
- **Basic functionalities:**
 - All platforms:
 - Secure access controls and encryption for protecting sensitive data.
 - Simple authentication methods to prevent unauthorized access.

Application Security Aspect

- Project planning and execution based on Secure Software Development Lifecycle (SSDLC).
- Application vulnerability assessment details, such as the tools to be used and the scope of the testing to be defined in the testing plan.
- Application vulnerability assessment results with the verification of the vulnerabilities.
- Recommendations and their implementation of the recommendations to address the vulnerabilities.

- Evidence of application vulnerability reassessment to verify that the vulnerabilities have been addressed successfully.

4. IoT-Enabled Plant Protection

- **Features:**
 - GPS and environmental sensors deployed near rare or endangered plant species.
 - Real-time data collection on location, temperature, humidity, and movement.
 - Automated alerts for unusual activity (e.g., potential poaching, habitat disturbance).
 - Centralized monitoring dashboard for administrators.
- **Outcome:**
 - Enhanced protection of endangered plant species through continuous monitoring.
 - Rapid response capability to environmental or security threats.
 - Long-term data records for habitat condition analysis.
- **Basic Functionalities:**
 - Real-time environmental sensing.
 - GPS-based plant and sensor tracking.
 - Threat detection and automated alerts.
 - Centralized monitoring dashboard.
 - Ecological data storage and analysis.

[Optional] Additional Innovation

1) Innovative Ground-Truthing Solution for Biodiversity Documentation:

- **Implementation:**
 - Develop a novel solution that enhances the existing biodiversity ground-truthing workflow by improving how plant species are identified, located, recorded, and managed in real-world forest environments.
- **Requirements:**
 - Identify limitations of current ground-truthing practices.
 - Propose and implement an improved workflow or system design.
 - Demonstrate how the proposed approach improves field data collection efficiency, accuracy, or scalability.
 - Evaluate the effectiveness of the proposed solution through suitable testing or validation.
- **Outcome:**
 - An innovative approach that advances biodiversity documentation beyond conventional methods.
 - Improved support for researchers and conservation officers in field-based biodiversity monitoring.
 - A scalable workflow suitable for protected forest ecosystems.